

# MGMT 405 – Managerial Economics

## Teaching Note – Module 2: Marginal Revenue

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### Preliminaries

We are building our toolkit to maximize profits in the generic case of a firm facing a downward sloping, linear demand curve. Such a firm has a simple decision to make: choose a spot on the demand curve, i.e. a combination of price charged and quantity sold. We have already seen the marginal-decision rule  $MR=MC$ : the firm should produce and sell a quantity  $Q$  such that its marginal revenue equals its marginal cost. This note focuses on the marginal revenue side. Marginal cost will be covered in Module 3. The two sides are brought together in the Teaching Note on  $MR = MC$ .

What is marginal revenue? In words, it is the change in revenues when you sell one more unit of the good. To get there, we start from total revenue. Let's write down a specific demand function (from the lecture slides; in practice you would estimate it from data, as in the Teaching Note on Regressions) and work through an example using this demand function:

$$Q = 1,600 - 4P$$

### The Recipe

We have to follow three steps to derive the equation for MR. You can apply this recipe whenever you face this type of problem (that will be often, throughout the MBA program!):

- 1) **Calculate inverse demand** (solve for  $P$ ).

$$P = 400 - \frac{1}{4}Q$$

(If you're wondering what I did here, I took our demand function, put  $P$  on the left side,  $Q$  on the right side and divided both sides by 4). This gives  $P$  as a function of  $Q$ , which we denote by  $P(Q)$ .

- 2) **Calculate total revenue** (multiply  $P(Q)$  by  $Q$ ). What's total revenue? It is  $P \times Q$ . So:

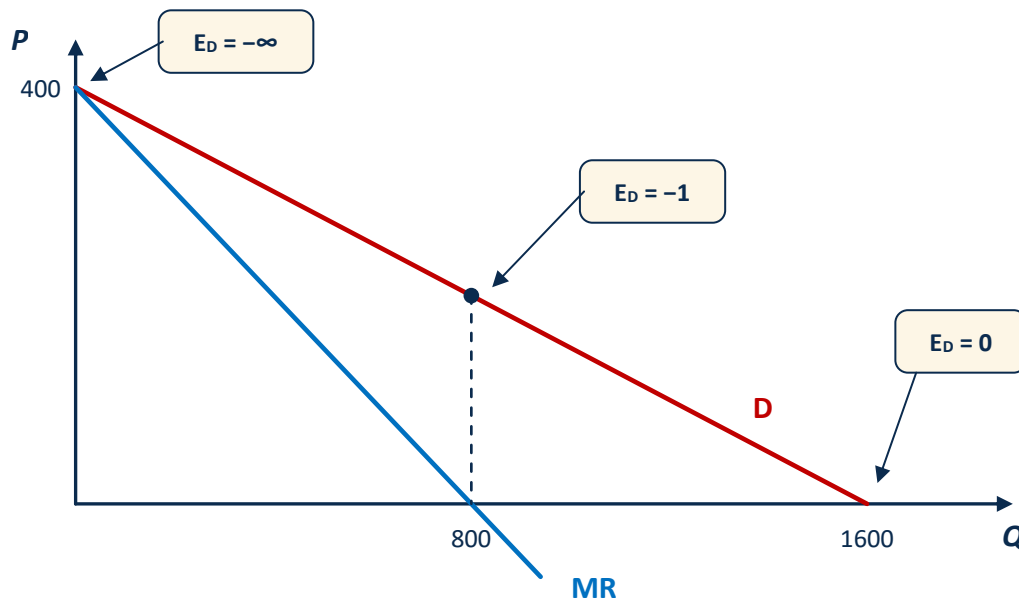
$$TR = P \cdot Q = 400 \cdot Q - \frac{1}{4}Q^2$$

(Here I just multiplied the previous equation by  $Q$ : easy!)

- 3) **Compute marginal revenue from total revenue** ( $MR = dTR/dQ$ ). That's the hardest step. It entails taking the derivative of  $TR$  with respect to  $Q$ . If you are unfamiliar with the underlying calculus, please watch the TA Math Review videos. All you need to know is that the derivative of  $Q$  with respect to  $Q$  is 1, and that the derivative of  $Q^2$  with respect to  $Q$  is  $2Q$ :

$$MR = \frac{dTR}{dQ} = 400 - \frac{1}{2}Q$$

So now we have MR. You will notice that MR has *twice the slope* of the inverse demand function. The slope of the inverse demand function is  $-1/4$ , and the slope of MR is  $-1/2$ . When demand is linear, MR is always twice as steep as (inverse) demand. Graphically, we obtain:



The figure also ties  $MR$  to the price elasticity of demand,  $E_D$ . On the upper half of the demand curve  $MR$  is positive and demand is elastic, so cutting the price raises total revenue. On the lower half  $MR$  is negative and demand is inelastic, so cutting the price lowers total revenue. The two halves meet where  $MR = 0$ . That is the unit-elastic point,  $E_D = -1$ , and it is where total revenue reaches its maximum. In our example  $MR = 400 - Q/2$ , so  $MR = 0$  at  $Q = 800$  – half of the 1,600 units that would be demanded at a price of zero.

We will elaborate on the features displayed in the figure in the Teaching Note on Demand Elasticity and Total Revenue.